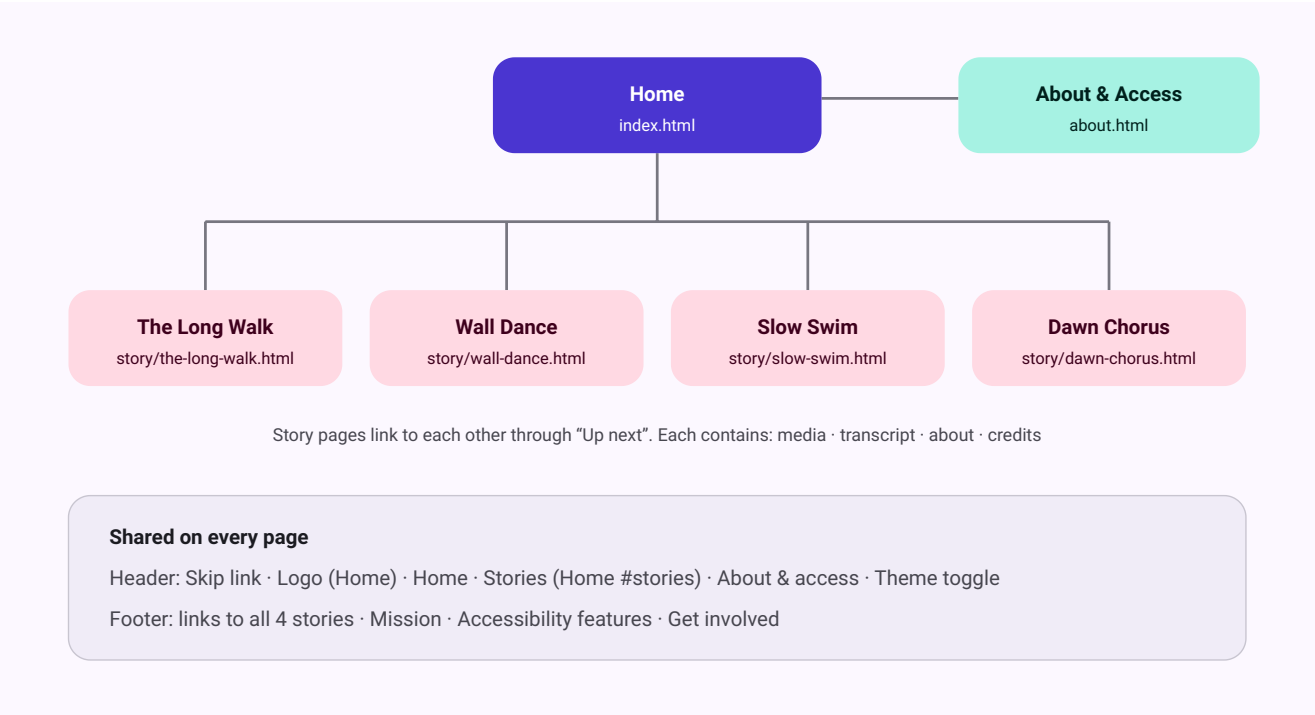
StoryCast: Everyday Voices

An accessible, three-page-type microsite for short audio and video stories told by ordinary people. It is built with semantic HTML5, Sass and vanilla JavaScript (no frameworks), styled after Google's Material 3 design system, and targets **WCAG 2.1 AA**.

Deliverables

Deliverable	Location
Research document (PDF)	`docs/research.pdf` (source: `docs/research.html`)
Sitemap (diagram)	`docs/sitemap.svg`
Wireframes (link)	Wireframes page on Figma: Home, Story and About & Access, desktop and mobile (also in the research PDF, section 6)
High-fidelity mockups (link)	StoryCast: Everyday Voices on Figma: all 6 pages at desktop (1440px) and mobile (390px)
Documentation (PDF)	`docs/documentation.pdf`
Codebase	this repository

Sitemap



Every page shares the same header (skip link, Home, Stories, About & access, theme toggle) and footer.

Project structure

```
storycast/
├── index.html
├── about.html
├── favicon.svg
├── story/
│   ├── the-long-walk.html
│   ├── wall-dance.html
│   ├── slow-swim.html
│   └── dawn-chorus.html
├── sass/
│   ├── main.scss
│   ├── abstracts/
│   ├── base/
│   ├── layout/
│   ├── components/
│   │
│   └── pages/
├── css/main.css
├── js/main.js
├── assets/
│   ├── media/
│   ├── captions/
│   ├── transcripts/
│   └── images/
└── docs/
```

CSS architecture

- **Cascade layers.** `sass/main.scss` declares `@layer reset, base, components, layout, pages;` and loads each folder into its layer with `meta.load-css()`. Later layers win, so a page or the app bar can adjust a component without out-specifying it.
- **Tokens.** Colours (`_colors.scss`, light and dark roles plus fixed media colours), type (`_typography.scss`) and spacing, shape, motion and breakpoints (`_spacing.scss`) are the only place raw values live. Partials read them through `color-role()`, `color-media()`, `space()`, `corner()` and the `up()` / `down()` breakpoint mixins.
- **BEM.** Every class is `block__element--modifier`; state uses ARIA attributes or `.is-*`. Bare element selectors are kept only for `.prose` (long-form text) and native <video> / <audio>.

Container query

The **media card** (`sass/components/_media-card.scss`) responds to the width of its own container, not the viewport:

Container width	Layout
under 18.5rem	Compact list row: small thumbnail beside the title (used for "Up next")

Container width	Layout
18.5rem to 40rem	Stacked card: 16:9 thumbnail above the text (used in the story grid)
40rem and up	Horizontal card with summary and access chips (used for the featured story)

The story grid on the home page is also a container (`card-grid`): it switches from 1 to 2 to 4 columns based on its own width. The "Up next" list (`card-stack`) is one column in the story sidebar, where its cards become list rows, and several columns of stacked cards when the sidebar drops below the transcript.

Running locally

Requirements: Node.js 18+.

```
npm install
npm run build
npm run serve
npm run watch
```

Use a local server rather than opening the files directly: browsers block caption files (`<track>`) on `file://` pages, and seeking needs a server that supports range requests (`http-server` does).

Rebuilding the PDFs

`docs/research.pdf` and `docs/documentation.pdf` are made from `docs/research.html` and `docs/documentation.html` with your installed Chrome. After editing either HTML file, rebuild them:

```
npm run docs:pdf
npm run docs:watch
```

`docs:watch` rebuilds a PDF every time you save its HTML file (Ctrl+C to stop). If Chrome isn't found, set `CHROME_PATH` to its location.

Accessibility checklist

Requirement	How it's met
Semantic structure	`header`, `nav`, `main`, `article`, `section`, `aside`, `figure`/`figcaption`, `footer`; one `h1` per page and a logical heading order
Skip link	"Skip to main content" is the first focusable element on every page
Captions	Every video and audio story has a WebVTT `<track kind="captions">`, on by default. The three video clips are silent, so their caption track says so
Transcripts	Every story has an interactive, timestamped transcript plus a `.txt` download. For the silent clips it describes what you see (WCAG 1.2.1)

Requirement	How it's met
Audio description	Every video has a `kind="descriptions"` track; the **Audio description** toggle reads it aloud with speech synthesis and lowers the video volume (falls back to an ARIA live region)
Contrast (1.4.3)	Every colour pairing in `_colors.scss` is ≥ 4.5:1 in light and dark themes (checked with a script)
Keyboard (2.1.1)	All controls are native `button`/`a`; ARIA tabs use arrow keys, Home and End; Esc closes the mobile menu
Shortcuts (2.1.4)	K, J, L, C and D only work while focus is inside the player
Focus visible (2.4.7)	A 3px two-tone focus ring on every interactive element
ARIA where needed	`aria-current` (nav, breadcrumb, transcript line), `aria-pressed` (toggles, filters), `aria-expanded`/`aria-controls` (menu), `role="tablist"`/`tab`/`tabpanel`, `role="status"` (filter results, form success)
Forms (3.3.1–3.3.3)	Visible labels, hints with `aria-describedby`, inline errors, `aria-invalid`, an error summary that receives focus
Reflow and zoom (1.4.10)	Single column at 320px; no horizontal scroll or overflowing content at 320, 480, 600, 768, 840, 1024, 1280 and 1920px
Motion (2.2.2, 2.3.3)	The home hero has a muted 10-second loop with a visible Pause/Play button. It doesn't start if `prefers-reduced-motion` is on (or without JS), and it has a text description. `prefers-reduced-motion` also disables transitions and smooth scrolling. No other media autoplays and nothing flashes
Themes	Follows the system theme; a toggle stores the choice; `forced-colors` (Windows High Contrast) is supported
Touch targets (2.5.8)	Interactive elements are at least 44 to 48px

Testing done: axe-core (WCAG 2.0/2.1 A + AA + best practice) reports 0 violations on all 6 pages in light and dark themes at 1440px and 390px. A scripted keyboard and interaction test covers tabs, menu, filters, player toggles, transcript seeking and form validation. The browser accessibility tree (what screen readers read) was reviewed on every page type: one `h1` and no skipped heading levels, header, navigation, main and footer landmarks on every page (each navigation named), no unnamed controls, and toggle states (`pressed`, `expanded`) exposed. **Screen readers:** manual results with Orca, NVDA and VoiceOver are recorded in Screen reader tests below.

Screen reader tests

Results of manual testing with real screen readers. Mark each cell **Pass**, **Issue** (with a note below the table) or leave it as **–** (not tested yet).

#	Page	Do this	You should hear	Orca + Firefox (Linux)	NVDA + Firefox (Windows)	VoiceOver + Safari (macOS / iOS)
1	Any page	Press Tab once	"Skip to main content, link"	–	–	–
2	Home	Press H repeatedly	The page title (h1), then "This week on StoryCast", "All stories" and the other section headings in order	–	–	–
3	Home	Move by landmark (Orca M, NVDA D, VoiceOver rotor → Landmarks)	Banner, "Main" navigation, main, the page sections, footer navigation, content info	–	–	–
4	Home	Move to the hero video button	"Pause background video, button"	–	–	–
5	Home	Activate the "Audio" filter	"Audio, toggle button, pressed", then "Showing 1 audio story."	–	–	–
6	Story	Move to the tabs, then press the arrow keys	"Transcript, tab, selected, 1 of 3", then the next tab	–	–	–
7	Story	Toggle "Captions"	"Captions, toggle button, pressed" / "not pressed"	–	–	–
8	Story	Turn on "Audio description" and play	Each description is spoken. Note whether it talks over the screen reader's own voice	–	–	–
9	Story	Activate a transcript timestamp	"Play from 24 seconds, button", then the video jumps there	–	–	–
10	Story	Use the native video or audio controls	Play, mute, time slider and full screen each have a name	–	–	–

#	Page	Do this	You should hear	Orca + Firefox (Linux)	NVDA + Firefox (Windows)	VoiceOver + Safari (macOS / iOS)
11	About	Submit the empty form	The error summary is read ("There are 5 problems"), and each field announces its error	–	–	–
12	About	Fill in the form correctly and submit	The success message is read	–	–	–
13	Any page, narrow window or 400% zoom	Open and close the menu, then press Esc	"Menu, button, collapsed" / "expanded"; focus returns to the menu button	–	–	–

Notes: *none yet*

Keys: **Orca** (Super + Alt + S to start; Orca key = Insert or Caps Lock; Orca + A switches browse/focus mode). **NVDA** (free from nvaccess.org, Windows only; Insert + Space switches browse/focus mode; Insert + F7 lists headings and links). **VoiceOver** (Cmd + F5 on a Mac; Settings → Accessibility → VoiceOver on iPhone; VO + U or a two-finger twist opens the rotor).

To test from another device, run `npm run serve` and open `http://<this-computer's-IP>:8080` (find it with `hostname -I`); from a VirtualBox VM use `http://10.0.2.2:8080`.

Media sources

The story videos and photos are linked directly from the Cloudinary sample library. The URLs use Cloudinary transformations (`c_limit,w_1280,q_auto` for video, `so_<seconds>` for poster frames, `c_fill` / `c_thumb,g_face` for photos), so each video loads at about 2 to 4.5 MB instead of 28 to 39 MB.

Where	Media
The Long Walk	`samples/elephants.mp4`
Wall Dance	`samples/cld-sample-video.mp4`
Slow Swim	`samples/sea-turtle.mp4`
Dawn Chorus cover	`samples/landscapes/beach-boat.jpg`
About: mission photo	`samples/people/jazz.jpg`
Home: call-to-action photo	`cld-sample.jpg`

The home hero uses a looping Gemini video (`assets/media/hero-loop.mp4`, poster `assets/images/hero-loop-poster.jpg`).

Credits

- Video clips and photos: Cloudinary sample library.
- **Dawn Chorus** audio is a StoryCast original, made with the Web Audio API.
- Fonts: Google Sans Flex and Material Symbols Rounded (Google Fonts).